Bachelor of Computer Applications (Hons. Game Development)
Semester- III

Subject: Database Management System

Assignment 8: Basic SQL Queries and Aggregate Functions

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1. Introduction

These queries run against the sample Library Management System dataset from Assignment 7.

1. Filter active members

Lists active members only.

```
SELECT member_id, name, membership_status
FROM Members
WHERE membership_status = 'Active'
ORDER BY member_id;
```

```
SELECT member_id, name, membership_status
FROM Members
WHERE membership_status = 'Active'
ORDER BY member_id;
```

RESULT GRID (illustrative output from the sample dataset)

member_id	name	membership_status
1	Aarav Sharma	Active
2	Diya Nair	Active
4	Meera Iyer	Active
6	Ananya Sen	Active

2. Sort books by publication year

Displays books from newest to oldest publication year.

```
SELECT title, publication_year
FROM Books
ORDER BY publication_year DESC;
```

```
SELECT title, publication_year
FROM Books
ORDER BY publication_year DESC;
```

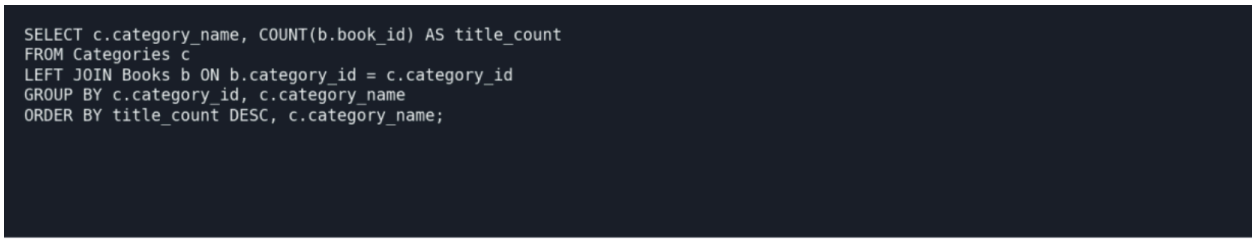
RESULT GRID (illustrative output from the sample dataset)

title	publication_year
Sapiens	2011
Wings of Fire	1999
Harry Potter and the Philosopher's Stone	1997
A Brief History of Time	1988
1984	1949
Animal Farm	1945
Malgudi Days	1943
Pride and Prejudice	1813

3. Count books in each category (GROUP BY)

Counts titles in each category, including categories with zero titles.

```
SELECT c.category_name, COUNT(b.book_id) AS title_count
FROM Categories c
LEFT JOIN Books b ON b.category_id = c.category_id
GROUP BY c.category_id, c.category_name
ORDER BY title_count DESC, c.category_name;
```



RESULT GRID (illustrative output from the sample dataset)

category_name	title_count
Fiction	3
Indian Literature	2
Fantasy	1
History	1
Science	1

4. Categories with at least two titles (HAVING)

HAVING filters grouped results after aggregation.

```
SELECT c.category_name, COUNT(b.book_id) AS title_count
FROM Categories c
JOIN Books b ON b.category_id = c.category_id
GROUP BY c.category_id, c.category_name
HAVING COUNT(b.book_id) &gt;= 2;
```

```
SELECT c.category_name, COUNT(b.book_id) AS title_count
FROM Categories c
JOIN Books b ON b.category_id = c.category_id
GROUP BY c.category_id, c.category_name
HAVING COUNT(b.book_id) >= 2;
```

RESULT GRID (illustrative output from the sample dataset)

category_name	title_count
Fiction	3
Indian Literature	2

5. Aggregate functions

Demonstrates COUNT, SUM, AVG, MIN and MAX on the eight sample book titles.

```
SELECT
  COUNT(*) AS total_titles,
  SUM(publication_year) AS sum_publication_years,
  ROUND(AVG(publication_year), 2) AS avg_publication_year,
  MIN(publication_year) AS oldest_publication_year,
  MAX(publication_year) AS newest_publication_year
FROM Books;
```

RESULT GRID (illustrative output from the sample dataset)

total_titles	sum_publication_years	avg_publication_year	oldest_publication_year	newest_publication_year
8	15645	1955.63	1813	2011

6. Available physical copies

Shows how many copies of each title are marked Available.

```
SELECT b.title, COUNT(*) AS available_copies
FROM BookCopies bc
JOIN Books b ON b.book_id = bc.book_id
WHERE bc.status = 'Available'
GROUP BY b.book_id, b.title
ORDER BY available_copies DESC, b.title;
```

```
SELECT b.title, COUNT(*) AS available_copies
FROM BookCopies bc
JOIN Books b ON b.book_id = bc.book_id
WHERE bc.status = 'Available'
GROUP BY b.book_id, b.title
ORDER BY available_copies DESC, b.title;
```

RESULT GRID (illustrative output from the sample dataset)

title	available_copies
Wings of Fire	2
1984	1
Animal Farm	1
Harry Potter and the Philosopher's Stone	1
Malgudi Days	0
Pride and Prejudice	1
Sapiens	1

2. Summary

WHERE filters individual rows, ORDER BY controls display order, GROUP BY forms groups for aggregation, and HAVING filters those groups. COUNT, SUM, AVG, MIN and MAX summarize values across the selected records.